2025 State Tournament - Div C -Georgia

Water Quality

Written by: Ryan Wood, Emma Beasley, Tori Kleinberg, Megan Walton, Selene Franklin

School Name/Team: ____

Student Names: ____

Instructions:

1. Write all answers on the Answer Sheet. DO NOT WRITE on Test Packet.
2. See the Image Sheet when asked to refer to an image by a question (it may be a separate sheet or attached to the back of this packet).
3. For numerical questions, answers within a reasonable range will be accepted.
4. When answering free-response questions, use the appropriate abbreviations for satellite and instrument names
5. For free-response questions, the answer sheet indicates the amount of space you should allocate to fully answer each question. Points are assigned based on accurately answering each part of the question.
6. Ties marked with a * will be broken in order

FOR GRADERS ONLY: ____________________ / 86 (Total points)

Part 1

1. Which of the following best describes what is broadly referred to as the water cycle? (1 pt)
    a. Movement of water between solid, liquid, and vapor states throughout the atmosphere and Earth.
    b. The cycle of liquid water throughout a given environment with respect to the organisms within it.
    c. The different ways in which an organism can utilize water chemically
    d. The cycle includes hydroxylation of an alkyl chain to form hydroxy compounds, which are used in organisms.
2. Percolation can have all of the following environmental effects except which one? (1 pt)
    a. Hypermineralization of a body of water
    b. Pollution of a nitrate into a body of water
    c. Increased runoff
    d. Algal bloom
3. Water possesses which of the following qualities? (1 pt)
    a. Hydrophobic
    b. Nonpolar
    c. Amphoteric
    d. Amphipathic
4. With regards to the bicarbonate buffer system in aquatic ecosystems, notably coral reefs, what would be expected if atmospheric carbon dioxide levels increased? (1 pt)
    a. Higher levels of bicarbonate lead to higher acidity in the water
    b. Higher levels of carbon dioxide are solvated and have no effect
    c. There are lower levels of dissolved carbon dioxide, creating a basic environment
    d. Higher levels of produced carbonic acid yield a lower pH
5. Which is not a step of potable water treatment? (1 pt)
    a. Coagulation
    b. Flocculation
    c. Sedimentation
    d. Nitration
    e. Filtration
6. What is the self-ionization constant of water at STP? (1 pt)

7. Match the following terms with their corresponding steps in the Nitrogen Cycle (4 pts)
    a. Denitrification: ______
    b. Nitrification: ______
    c. Ammonification: ______
    d. Nitrogen Fixation: _______

    1. Conversion of N2 to NH3
    2. Conversion of NO3- to N2
    3. Conversion of N to NH3
    4. Conversion of NH3 to NO3-

8. Sewage waste, if released untreated, can introduce numerous harmful organisms into water bodies causing significant pollution. From the list below select all the types of organisms emitted from sewage waste that are known to cause water pollution. (4 pts)
    a. Algae
    b. Plastics
    c. Bacteria
    d. Helminths
    e. Insects
    f. Protozoa
    g. Plankton
    h. Fungi

9. An upstream watershed has experienced deforestation to create farmland for a nearby town. How would this land usage change impact the aquatic ecosystem downstream? Include nutrient cycling, macroinvertebrate diversity, and sedimentation in your answer. (4 pts)

10. Define the term riparian buffer and include two examples of how riparian buffers help protect watersheds. Be specific in your examples. (3 pts)

11. Two lakes receive the same amount of acid rain. Lake A lies on granite bedrock, while Lake B lies on limestone bedrock. Which lake is more vulnerable to acidification and why? Explain why the other lake is more resistant. (*3 pts)

12. Within the multistage sequence of wastewater treatment, what is the unique role performed by the sedimentation process? (1 pt)
    a. It is a chemical process that neutralizes the electrical charge of suspended particles allowing them to aggregate before being physically mixed.
    b. It is a process that uses gravity to separate and remove large, dense suspended solids.
    c. It is a process that removes dissolved organic compounds that interfere with the final disinfection stage, making the process more efficient.
    d. It is a biological process where microorganisms consume organic matter and settle.

13. Define the process of eutrophication and explain how it impacts an aquatic ecosystem. Be sure to include nutrients(types), algae, decomposition, oxygen, and hypoxia in your answer. (4 pts)

14. In a typical aquatic food web, what is the crucial role of phytoplankton and why are they considered the foundation of most marine ecosystems? (2 pts)

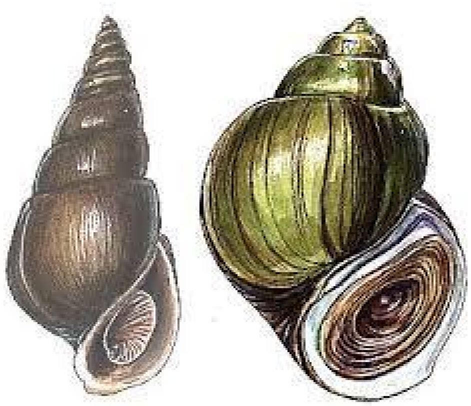
15. **Fill in the blank**: Increased atmospheric CO2 directly ____________ (increases or decreases) oceanic pH, resulting in a critical challenge for calcifying organisms is the resulting decrease in the mineral's saturation state. This is especially affecting calcifying organisms that use the metastable polymorph of calcium carbonate which is known as _______________. This makes the biomineralization process thermodynamically ____ ____________ (favorable or unfavorable), hindering shell formation long before the water becomes corrosive enough to cause _________________. (2 pts)

Part 2

16. Identify the following species by common name: (1 pt)

17. Identify the following species by common name: (1 pt)

18. Identify the following species by common name: (1 pt)



19. Identify the following species by common name: (1 pt)

20. Identify the following species by common name: (1 pt)

21. **Match the organism to the correct class (there might be more than one organism to each class):** (7 pt)

a. Crane Fly	Pollution Sensitive (c and d)
b. Water Boatman	Moderately Tolerant (e)
c. Stonefly	Air Breathing (b)
d. Gilled Snails	Pollution Tolerant (f)
e. Flatworm	Moderately Sensitive (a and g)
f. Tubifex	
g. Scuds	

22. What is the proper diet for leeches? (2 pt)

23. Explain why Mayfly, Caddisfly, and Stonefly are reliable bioindicators for water quality? (*3 pt)

24. The zebra mussel can be classified as what type of animal? Describe an ecological consequence of introducing the zebra mussel into a new freshwater ecosystem. Be specific in your answer (3 pt)

25. Which of the following macroinvertebrates are predators during their aquatic larval stages? *Select all that apply: (4 pt)*
   a. Dragonfly
   b. Mayfly
   c. Water Boatman
   d. Predacious Diving Beetle
   e. Water strider
   f. Damselfly

26. Unlike the complete metamorphosis of insects like mosquitoes, stoneflies undergo ______________________ metamorphosis, where the immature aquatic stages resemble the adult form without developing functional ______________. (2 pt)

27. Asian carp, an ______________ ____________ (two words) species, are especially problematic because their filter feeding habit allows them to outcompete native species by removing large quantities of ______________ from the water. (2 pt)

28. Which organisms are likely to increase in abundance when sedimentation rates rise and benthic oxygen levels fall? *Select all that apply: (3 pt)*
    a. Mayfly
    b. Tubifex worm
    c. Blood midge
    d. Stonefly
    e. Aquatic sowbug
    f. Deer/Horse fly larva
29. Compare the development time, reproductive output, and habitat needs of Class 1 organisms with Class 4 organisms. How do these traits relate to their ability to colonize disturbed habitats? (4 pt)

30. A sample from a single spot in a slow- moving densely vegetated stream reveals a healthy population of Gilled Snails living alongside a large population of air-breathing snails. What does the presence of both organisms reveal about the stream's environment? (1 pt)
    a. The water has consistently high levels of dissolved oxygen.
    b. The stream is heavily polluted with toxic chemicals.
    c. The dissolved oxygen levels likely fluctuate dramatically, dropping very low at night.
    d. The gilled snails are more recent, invasive species, displacing the native air breathing snails.

Part 3

31. The hydrogen ion concentration for a given solution is [H+] = 1.73 * 10^3 M. What is the pH? (1 pt)

32. Why are phosphates hard to measure? Name one common method of measuring it.(*2 pt)

33. How many measurements are taken to assess biological oxygen demand? If less than 5 measurements, at what frequency are the tests typically spaced out? (2 pt)

34. What is an approximate normal level for turbidity dissolved O2 (with units)? (1 pt)

35. A water sample taken downstream from a wastewater treatment plant shows a Biochemical Oxygen Demand (BOD) of 50mg/L and a Dissolved Oxygen (DO) level of 2mg/L. A sample taken upstream showed a BOD of 3mg/L and a DO of 8mg/L. What is the most direct conclusion from the given data? (1 pt)
    a. The plant is discharging water with high levels of toxic metals
    b. The plant is likely releasing insufficiently treated sewage with high organic matter content.
    c. The water temperature has significantly increased, causing the drop in DO.
    d. The stream is experiencing a natural algal bloom unrelated to the treatment plant.
36. A coastal monitoring station detects a sudden drop in salinity after a major storm. Which of the following would you expect to see a sharp increase in? (1 pt)
    a. Fecal Coliform and pH
    b. Water temperature only
    c. pH and dissolved oxygen
    d. Turbidity and Nitrates
37. A water quality report lists a pH of 4.5 for a lake. What is the primary environmental concern associated with this pH level? (1 pt)

38. A standard Biochemical Oxygen (BOD) test is performed on a water sample:
    a. Write the simple mathematical equation used to calculate the BOD values from the test measurements: (1 pt)
    b. Define what each term stands for in your equation (MUST HAVE UNITS): (2 pt)
    c. Explain what a high BOD value signifies about the water quality: (1 pt)

39. To measure fecal coliform, a lab technician must dilute a water sample. They mix 1mL of sample with 99mL of sterile water. They then plate 1mL of this diluted mixture and count 32 colonies after incubation. What is the reported fecal coliform count for the original, undiluted sample? (1 pt)

40. A stream with a population of trout experiences a temperature increase from 15^0C to 25^0C due to the removal of stream side vegetation. How does this change affect the dissolved oxygen (DO) and the metabolic rate and oxygen demand of the trout? (*2 pt)

41. A 100 mL water sample requires 3.50 mL of 0.0141 N $AgNO_3$ in a chloride titration. What is the chloride concentration in mg/L? (1 pt)